

SKILL - 4

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JupyterLab Python 3 (jupyterlab)

[4]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge, Lasso, ElasticNet
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# Load dataset
housing = fetch_california_housing(as_frame=True)
df = housing.frame

# Separate X and y
X = df.drop("MedHouseVal", axis=1)
y = df["MedHouseVal"]

# Split data: 80% training, 20% testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training data:", X_train.shape)

# Linear Regression
lr = LinearRegression()
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)

# Evaluation
mse_lr = mean_squared_error(y_test, y_pred_lr)
rmse_lr = np.sqrt(mse_lr)
mae_lr = mean_absolute_error(y_test, y_pred_lr)
r2_lr = r2_score(y_test, y_pred_lr)

print("\nLinear Regression")
print("RMSE:", rmse_lr)
print("MAE:", mae_lr)
print("R2:", r2_lr)

# Coefficients
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": lr.coef_
})

print("\nCoefficients:")
print(coefficients)
```

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```
# Standardization
scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
print("\nScaling completed successfully.")
```

Training data: (16512, 8)

Linear Regression
 RMSE: 0.7455813830127761
 MAE: 0.5332001304956558
 R2: 0.5757877060324511

Coefficients:

	Feature	Coefficient
0	MedInc	0.448675
1	HouseAge	0.009724
2	AveRooms	-0.123323
3	AveBedrms	0.783145
4	Population	-0.000002
5	AveOccup	-0.003526
6	Latitude	-0.419792
7	Longitude	-0.433708

Scaling completed successfully.